POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2023
Programme: BE | Full Marks: 100
Course: Problem Solving Techniques (New) | Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) In a PST class there are K students. At the beginning of each class hour, each student shakes hands with each of the other students. If K is even, then will the number of handshakes that takes place be even or odd? If K is odd, then will the number of handshakes that takes place be even or odd? 8

   b) Derive the sum of first k square positive integer. 7

2. a) Explain why it is impossible to have a polyhedron with 6 triangular faces meeting at each vertex? 8

   b) The perimeter of certain right triangle is 60 inches. The height perpendicular to the hypotenuse is 12 inches. What is the length of the three sides of the triangle? 7

OR

Develop an expression as a formula to find out area of a triangle using only the lengths of the triangle's three sides.

3. a) A ten foot pole is dropped into a milling saw and randomly cut into three shorter poles. What is the probability that these three pieces will form a triangle? 7

   b) Suppose that you have 9 pearls. They all look the same, but 8 of have equal 8 weight and one is different. The odd pearl is either heavier or lighter; you do not know which. The only equipment that you have at hand is a balance scale. How can you use the scale to find the odd pearl in just three weighing? 8

4. a) A game is played by two players. They begin with a pile of thirty chips, all the same. For his or her move, a player may remove 1 to 6 chips. The player who removes the last chip wins. What strategy can the first player use so that he will always win? 7

OR

Solve the crypto arithmetic problem. ***CROSS + ROADS = DANGER*** 8

   b) Six people, named A, B, C, D, E, F are in the dining car of a train. They are one each from New York City, Chicago, Tulsa, St. Louis, Milwaukee and Atlanta. The following facts are known.

i. A and the man from New York City are physicians.
ii. E and the woman from Chicago are teachers.
iii. The person from Tulsa and C are engineers.
iv. B and F are veterans of Gulf war, but the person from Tulsa has never served in the military.
v. The person from Milwaukee is older than A.
vi. The person from Atlanta is older than C
vii. At St. Louis, B and the man from New York get off.
viii. At San Francisco, C and the man from Milwaukee gets off.

Match the names of the people with their professions and their cities.

5. a) There are two married couples that need to cross a river. A small boat is available that will hold just two people at a time. The males involved are quite jealous. No woman can be left with a man unless her husband is also present. There are no other constraints. How can these four people cross the river? What is the fewest numbers of trips possible? 8

b) A martini is made is made by mixing k parts gin with 1 part vermouth. Gin is usually 40% alcohol while vermouth is 20% alcohol. A martini is said to be dry if it contains relatively little vermouth. For instance, if k = 15 then the martini is said to be dry. If instead k = 5, then the martini is said to be sweet. Discuss by calculating the amount of alcohol in a dry martini vs sweet martini, which one is better? 7

6. a) A certain number k is a multiple of 9. Add the digits together. If the result has more than one digit, add those together. Continue adding digits together until you have a one digit answer. It will be a 9. Explain why this is so? 8

b) A cube of a side r is inscribed in a sphere. The sphere is inscribed in a cone with side length equal to the diameter of its base. The cone is inscribed in a right circular cylinder. What is the surface area of the cylinder? 7

7. Write short notes on: **(Any two)** 2×5

a) Explain Impossible Problems.

b) The following sequence has become known as the John H. Conway sequence. Determine the next term of the sequence.
1,1,1,3,1,4,1,1,3,6,1,2,3,1,4,8,1,3,3,2,4,1,6,?

c) Which is greater $sin(\cos x)$ or $cos(\sin x)$?